Technical Document #434: Retrieval Augmented Generation - Comprehensive Overvi

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Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Hybrid search combines keyword-based and semantic search.
- Semantic search finds documents based on meaning rather than exact keywords.